

Gauge and Homological Structures in Quantum Error Correction



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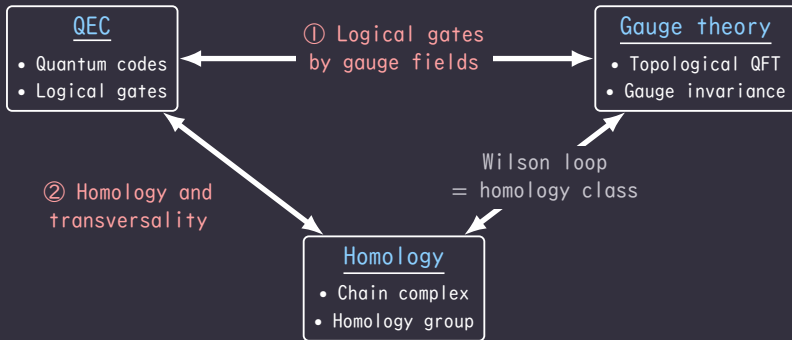
Kyoto University

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Based on arXiv:2511.15224, 2602.14499

Gauge – QEC – Homology, a shared backbone



Claim 1. Logical gates on any CSS code = exponentials of lattice gauge fields on its chain complex. → Part II

Claim 2. Whether a logical Z-rotation is transversal is decided by homology with $\mathbb{Z}_2$ coefficients. → Part III

Talk plan

- Preliminary Part I
- Logical gates by gauge field formalism Part II
- Homological origin of transversality Part III
- Summary Part IV

What is Quantum Error Correction?

- Logical info $\hookrightarrow$ larger Hilbert space
- Quantum (CSS) code: X/Z -type commutative “stabilizers”

$$S^X = \prod_{j=1}^n (X_j)^{c_j} \quad \text{or} \quad S^Z = \prod_{j=1}^n (Z_j)^{c_j}$$

- Logical state $|\psi_L\rangle$: fixed by every stabilizer

$$\forall S \quad S|\psi_L\rangle = +1|\psi_L\rangle$$

For high-energy physicists:

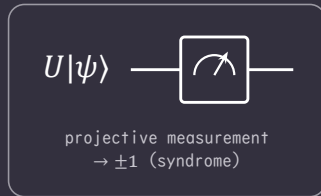
(X -)stabilizers = Gauss law op. Code space = gauge-invariant subspace

How to detect and correct?

- Measurement of stabilizers $\Rightarrow$ Error detection

$$|\psi_L\rangle \xrightarrow{\text{error}} U|\psi_L\rangle \xrightarrow{\text{measure}} S = \pm 1$$

- Flipped result (= “syndrome”)
 $\Rightarrow$ Error occurred!
- Error-correcting capability
= Min. weight of errors that
do not flip syndromes

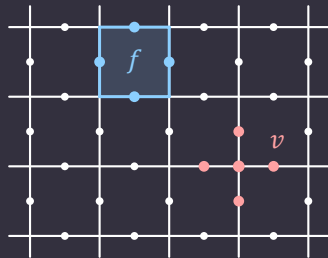


$\mathbb{Z}_2$ lattice gauge theory

- Quantum code = topological $\mathbb{Z}_2$ lattice gauge theory
- Kogut-Susskind Hamiltonian on 2D torus

$$H = - \sum_v A_v - \sum_p B_p$$

- $A_v = \prod_{e \ni v} X_e$ (Gauss law)
- $B_p = \prod_{e \in \partial p} Z_e$ (Flux)
- Z_e = link variable (gauge field)



$\mathbb{Z}_2$ lattice gauge theory as quantum codes

- Dictionary

Gauss law A_p $\leftrightarrow$ X stabilizer

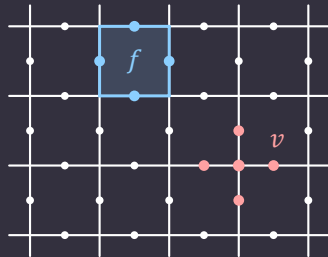
Flux B_p $\leftrightarrow$ Z stabilizer

link variable $Z_e \leftrightarrow$ qubit

- Ground states = logical state

$$A_v|\psi\rangle = B_p|\psi\rangle = +1|\psi\rangle$$

- Ground-state degeneracy = dimension of **logical space**

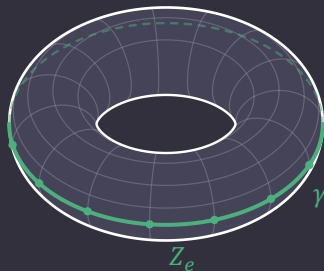


Wilson loop

- How many logical qubits = ground states?
- **Wilson loop** on non-contractible loop γ

$$W(\gamma) := \prod_{e \in \gamma} Z_e$$

- Gauge-invariant
= commute with Gauss law op.
 $\Rightarrow$ **physical observable**



Logical gates beyond Pauli

- Z : phase flip, X : bit flip $\Rightarrow$ not enough for universal quantum computation
- Solovay-Kitaev theorem: Full Clifford + 1 non-Clifford is enough

generating gate set

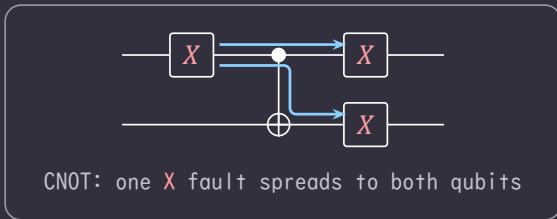
$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & \\ & i \end{pmatrix}, \quad CZ = \text{diag}(1, 1, 1, -1), \quad T = \begin{pmatrix} 1 & \\ & e^{i\pi/4} \end{pmatrix}$$

Problem 1

How to construct logical gates
beyond Pauli in general?

Need for transversal gates

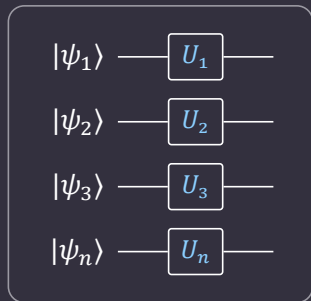
- Multi-qubit gates **amplify** error



- Destroys error-correction capability
- What kind of implementation is better? → “**Transversal gates**”

Transversal gate

- An independent single-qubit unitary on every qubit
- Do not amplify error $\Rightarrow$ **fault-tolerant**
- Limited class of unitaries $\rightarrow$ Q.
Universal? **No**



When and which transversal gates?

- **Eastin-Knill theorem** [Phys. Rev. Lett. 102, 110502 (2009)]
 - No universal gate set only with transversal gates
 - ⇒ Limits **what** is possible, not **which** gates for a given code
- Many examples, but **no unifying rule**

Problem 2

Which code can implement **which** gate transversally?

Two open problems

Problem 1

How to construct logical gates beyond Pauli in general?

⇒ Part II [arXiv:2511.15224](#)

Problem 2

Which code can implement which gate transversally?

⇒ Part III [arXiv:2602.14499](#)

Logical gates by gauge field formalism

Junichi Haruna
arXiv:2511.15224

Lattice gauge theory $\leftrightarrow$ QEC dictionary

Lattice gauge theory	Quantum error correction
Gauss-law generator	X -stabilizer S^X
Magnetic flux / curvature	Z -stabilizer S^Z
(Gauge-invariant) ground states	Code space $\mathcal{H}_c$
Wilson loop $W(\gamma) = \exp(i \oint_{\gamma} a)$	Logical $\bar{Z}(\gamma)$

Key questions

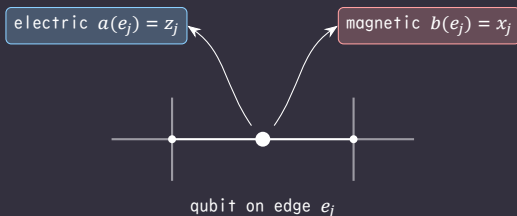
- How to construct logical gates **beyond Pauli** in general?
- What corresponds to logical gates in **gauge theory**?

Gauge field formalism

- Gauge field in lattice gauge theory = linear map from qubit (= edge) to operator
- Each qubit $e_j \rightarrow$ projectors onto $|1\rangle_j$ or $|-\rangle_j$:

$$a(e_j) = z_j = (I - Z_j)/2 = |1\rangle\langle 1|_j \quad \text{electric}$$

$$b(e_j) = x_j = (I - X_j)/2 = |-\rangle\langle -|_j \quad \text{magnetic}$$



Recovering gates from gauge fields

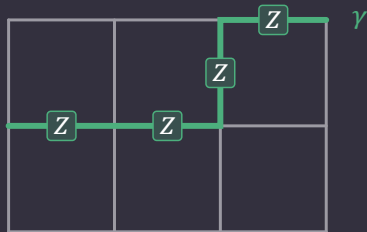
- Single-qubit gates come straight back:

$$Z_j = \exp(i\pi a(e_j)),$$

$$X_j = \exp(i\pi b(e_j))$$

- Transversal Z gates

$$\prod_{i=1}^n (Z_i)^{\gamma_i} = \exp(i\pi a(\gamma))$$



Stabilizers & logical operators

- Stabilizers from the (co)boundary maps:

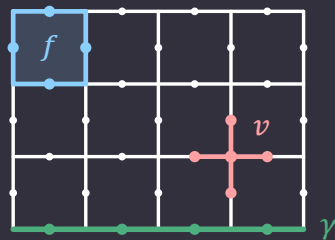
$$S_f^Z = \exp(i\pi a(\partial f)),$$

$$S_v^X = \exp(i\pi b(\partial^T v))$$

- Logical Z operator = Wilson loop:

$$\bar{Z}(\gamma) = \exp(i\pi a(\gamma))$$

$$\leftrightarrow W(\gamma) = \exp\left(i \oint_{\gamma} a\right)$$



Logical operators = gauge invariant

- Logical operator preserves the code space:

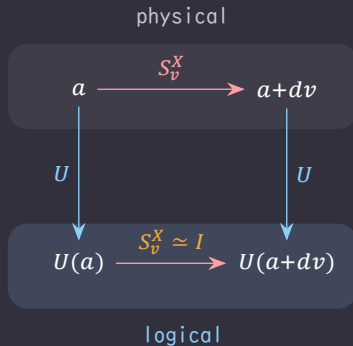
$$S^\dagger U S \simeq U \text{ on } \mathcal{H}_c$$

- X -stabilizer conjugation = gauge transformation

$$(S_v^X)^\dagger a S_v^X = a + dv \pmod{2}$$

- Logical = gauge-invariant:

$$\begin{aligned} (S_v^X)^\dagger U(a) S_v^X &= U(a + dv) \\ &\simeq U(a) \end{aligned}$$



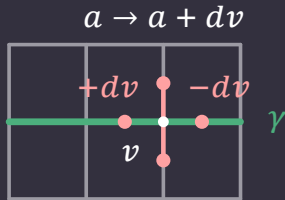
Construction principle

- Any function $U(a(\gamma))$ is gauge invariant:

$$a(\gamma) = \oint_{\gamma} a \rightarrow \oint_{\gamma} (a + dv) = \oint_{\gamma} a$$

- Which gate? → Replacement principle
- Replace physical X, Z with logical $\bar{X}, \bar{Z}$:

$$\mathcal{O}(X, Z) \rightarrow \overline{\mathcal{O}}(\overline{X}, \overline{Z})$$



Building logical S ($n=3$)

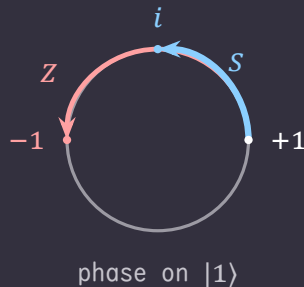
- S gate

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \exp\left(\frac{i\pi}{2} \frac{I - Z}{2}\right)$$

- Replacement principle: $Z \rightarrow \bar{Z}(\gamma)$ to get logical S

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi}{2} \frac{I - \bar{Z}(\gamma)}{2}\right)$$

- How to decompose $\bar{Z}(\gamma)$ into gauge fields?



Expanding $\bar{Z}(\gamma)$ in gauge fields

- $\bar{Z}(\gamma)$ as polynomial of projectors

$$\bar{Z}(\gamma) = Z_1 Z_2 Z_3 = (I - 2z_1)(I - 2z_2)(I - 2z_3)$$

- Relation between projectors and gauge fields:

$$z_1 + z_2 + z_3 = a(\gamma),$$

$$z_1z_2 + z_1z_3 + z_2z_3 = \frac{1}{2} \left(a(\gamma)^2 - a(\gamma) \right),$$

$$z_1 z_2 z_3 = \frac{1}{6} (a(\gamma)^3 - 3a(\gamma)^2 + 2a(\gamma))$$

- $\bar{Z}(\gamma)$ as a polynomial of $a(\gamma)$:

$$\bar{Z}(\gamma) = I - \frac{20}{3}a(\gamma) + 6a(\gamma)^2 - \frac{4}{3}a(\gamma)^3$$

Simplifying $\bar{S}(\gamma)$ modulo 4

- $\bar{S}(\gamma)$ as a polynomial of gauge fields:

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi}{2}\left(\frac{10}{3}a(\gamma) - 3a(\gamma)^2 + \frac{2}{3}a(\gamma)^3\right)\right)$$

- Simplified modulo 4 (since $S^4 = I$):

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi}{2}a(\gamma)^2\right)$$

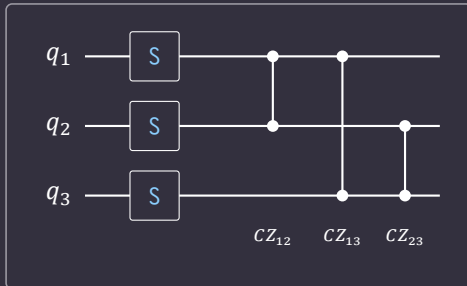
Physical gate decomposition ($n = 3$)

- Expansion of gauge fields polynomial

$$\begin{aligned}\bar{S}(\gamma) &= \exp\left(\frac{i\pi}{2}(z_1 + z_2 + z_3)^2\right) \\ &= \exp\left(\frac{i\pi}{2}(z_1 + z_2 + z_3)\right) \\ &\quad \times \exp(i\pi(z_1z_2 + z_1z_3 + z_2z_3))\end{aligned}$$

- First factor = S on each qubit,
second factor = CZ on every pair

$$\bar{S}(\gamma) = S_1 S_2 S_3 \cdot CZ_{12} CZ_{13} CZ_{23}$$



Other logical gates

- This strategy works for various logical gates

Logical gate	Gauge field expression
$\bar{S}(\gamma)$	$\exp(i\pi a(\gamma)^2/2)$
$\bar{H}(\gamma)$	$\bar{S}(\gamma) \exp(i\pi b(\tilde{\gamma})^2/2) \bar{S}(\gamma)$
$\overline{CZ}(\gamma_1, \gamma_2)$	$\exp(i\pi a(\gamma_1)a(\gamma_2))$
$\bar{T}(\gamma)$	$\exp(i\pi(2a^3 - 3a^2 + 2a)/4)$

- No special structure on QEC codes: applicable for general CSS codes

Summary of Part II

- Dictionary: $\mathbb{Z}_2$ lattice gauge theory $\Leftrightarrow$ quantum codes
- Systematic construction:
 - Logical gate = exponential of a polynomial of gauge fields
 - Replacement principle: $Z \rightarrow \bar{Z}$, $X \rightarrow \bar{X}$
 - Applicable to general quantum (CSS) codes

These gates are neither transversal, nor fault-tolerant in general.
When is a logical gate transversal? $\rightarrow$ Part III

Homological origin of transversal implementability

Junichi Haruna
arXiv:2602.14499

Motivation

- Transversal gates $\Rightarrow$ **fault-tolerant**, but *which* for **which** code?
- Many explicit constructions:
 - Triorthogonal codes, 2^m -divisible codes, quantum Reed-Muller codes...
- Gauge field construction $\Rightarrow$ not transversal

Key questions

- When a transversal gate gives a *logical* gate?
- Is there *one* principle behind these criteria?

Transversal Z and S rotations

- Transversal Pauli Z -rotations:

$$U(\theta/2) := \prod_{j=1}^n \exp\left(i\pi \frac{\theta_j}{2} \frac{I - Z_j}{2}\right) = \prod_{j=1}^n (S_j)^{\theta_j} \quad \theta \in \mathbb{Z}_4^n$$

- $S = \sqrt{Z}$ is the $\pi/2$ rotation, and $S^2 = Z$
- Goal: when does a transversal S realize the logical $\sqrt{Z}(\gamma)$?



When does a transversal logical S exist?

- Stabilizers fix the code structure
- $S = \pi/2$ rotation of Z
- $\Rightarrow$ Decidable only from the stabilizers and $\overline{Z}$

Result (preview)

- Transversal implementability of $S \Leftrightarrow$
two obstruction maps vanish for $\overline{Z}(\gamma)$; $\beta_1([\gamma]) = \beta_2([\gamma]) = 0$

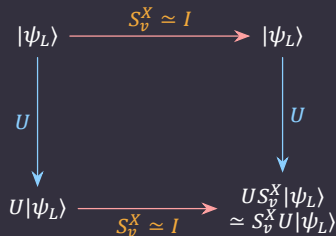
Logicality from the commutation relation

- Logicality $\Leftrightarrow$ commutativity on $\mathcal{H}_c$
- Commutation relation with X -stabilizer:

$$S^X(h)^\dagger U^\dagger\left(\frac{\theta}{2}\right) S^X(h) U\left(\frac{\theta}{2}\right) \\ = \underbrace{e^{i\pi\langle\theta,h\rangle/2}}_{\text{const. phase}} \underbrace{U(\theta \circ h)}_{\text{Z-string}}$$

 $\theta \circ h$: entrywise product

- Z-string must coincide with a stabilizer (and phase trivial)



$$S^\dagger U^\dagger S U \simeq I \quad \text{on } \mathcal{H}_c$$

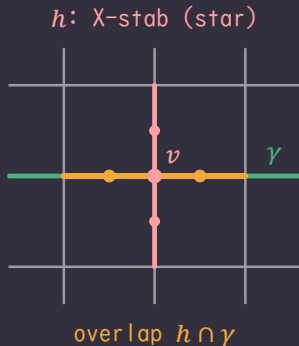
Two logicality conditions on θ

- Two conditions on θ , different moduli:

1. Phase factor: $H\theta = 0 \pmod{4}$

2. Z-string = stabilizer ($\perp h, \gamma$):
 $(H \odot K)\theta = 0 \pmod{2}$

- Which logical Z ? $\rightarrow$ further ($S^2 = Z$) needed



H = stack of $\{h\}$, K = stack of $\{h, \gamma\}$; $\odot$: row-wise product

Fixing the logical Z via $S^2=Z$

- How does θ relate to $\bar{Z}(\gamma)$?
- Constraint: square of $U(\theta/2)$ must be $\bar{Z}(\gamma)$

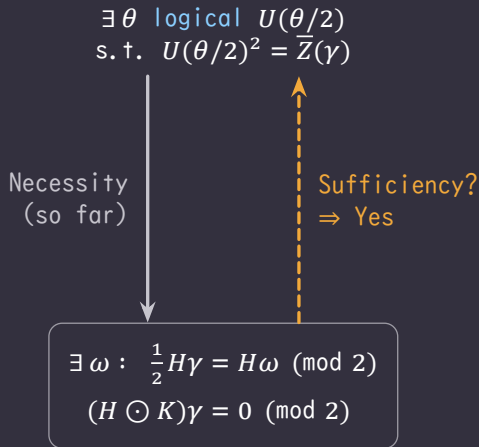
$$U(\theta/2)^2 = \prod_i [(S_i)^{\theta_i}]^2 = \prod_i Z_i^{\theta_i} = \bar{Z}(\gamma)$$

- Additional constraint on θ with some $\mathbb{Z}_2$ vector ω

$$\theta = \gamma \pmod{2} \iff \theta = \gamma + 2\omega \pmod{4}$$

Two logicality conditions on γ

- Two conditions on γ :
 1. $\exists \omega, \frac{1}{2}H\gamma = H\omega \pmod{2}$
 2. $(H \odot K)\gamma = 0 \pmod{2}$
- Necessary for transversal logical S
- Sufficient? $\Rightarrow$ Yes (next)



Main theorem

Theorem. Transversal implementability of logical S gate
 $\Leftrightarrow$ vanishing of two obstruction maps for $\bar{Z}(\gamma)$:

$$\beta_1([\gamma]) = \beta_2([\gamma]) = 0$$

- $\beta_1 := [(H \odot K)\gamma] \in \mathbb{Z}_2^n$: additional condition mod 2
- $\beta_2 := [H\gamma/2] \in \mathbb{Z}_2^n / \text{im } H$: liftability condition modulo 4

$\beta_2 =$ Bockstein obstruction

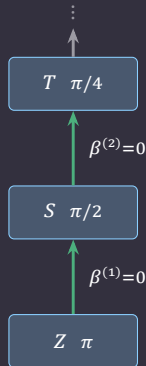
- $Z \rightarrow S$: halving the angle $\pi \rightarrow \pi/2$ = lifting the coefficients $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$
- β_2 measures its obstruction
- Transversal implementability = Existence of lift

$$\begin{array}{ccccccc}
 & [H\gamma/2] & & [\theta] & & [\gamma] & \\
 0 & \longrightarrow & \mathbb{Z}_2 & \xrightarrow{\times 2} & \mathbb{Z}_4 & \xrightarrow{\text{mod } 2} & \mathbb{Z}_2 \longrightarrow 0 \\
 & & \uparrow & & & & \uparrow \\
 & & & & & &
 \end{array}$$

β_2 : Bockstein connecting map
keeping $[\gamma] = [\theta \bmod 2]$

Beyond Z and S

- Same argument for climbing the **Clifford hierarchy**: $Z \rightarrow S \rightarrow T \rightarrow \dots$
- Transversal $\Leftrightarrow$ a **ladder of vanishing obstructions** $\beta_1^{(v)} = \beta_2^{(v)} = 0$
- Known criteria (divisibility, tri-orth.)
 $\Leftarrow \beta=0$ with uniform θ



Summary of Part III

- Transversal logical $S \Leftrightarrow \beta_1([\gamma]) = \beta_2([\gamma]) = 0$ (two obstruction maps vanish)
- The β maps are **homology maps**; β_2 = Bockstein-type lifting obstruction
- One decidable test: unifies the zoo, climbs the Clifford hierarchy

Transversal implementability of diagonal gates = a homological lifting obstruction.

Summary and
Future directions

Summary

Gauge theory · quantum error correction · homology theory
share common mathematical structures.

- Logical gates: built systematically from lattice gauge fields
- Transversality: governed by homology beyond $\mathbb{Z}_2$
- Their interplay $\rightarrow$ new, unifying insight

Future directions

- **Fault-tolerance** written in the language of lattice gauge theory / homology.
- **Non-Abelian extension**: non-Abelian codes & the quantum double model (Kitaev).
- Higher cohomological invariants and non-Clifford gates.

A shared language, still mostly unexplored.

Thank you!

Backup

Steane $[[7, 1, 3]]$: transversal S yes, T no

- Self-dual CSS ($H_X = H_Z = H$, the $[7, 4, 3]$ Hamming code): a single logical class

gate	lift	obstruction	transversal?
$S(\pi/2)$	$1 \rightarrow 2$	$\beta_1^{(1)} = \beta_2^{(1)} = 0$	yes
$T(\pi/4)$	$2 \rightarrow 3$	$\beta_2^{(2)} \neq 0$	no

- S : lift $\hat{\theta} = 3 \cdot 1_7$ gives $\langle \hat{\theta}, 1_7 \rangle = 1 \pmod{4}$, so $\prod_i S_i^\dagger$ is logical S
- T : every logical $\theta^{(3)} = 2c_0 1_7 + 4 \sum_i c_i h_i$ is even, so $\langle \theta^{(3)}, 1_7 \rangle \neq 1 \pmod{8}$: no transversal T , even for non-uniform $\theta \in \mathbb{Z}_8^7$
- $[[15, 1, 3]]$ Reed-Muller: transversal T (backup).

The obstruction detects the absence of transversal T : Eastin-Knill, from obstruction theory.

Cohomology dependence

- Logical action depends only on the **cohomology class**
- Gauge transformation $a(\gamma) \rightarrow a(\gamma) + dv$: Logical operator = Gauge-invariant operator

$$U(\gamma) \simeq U(\gamma + \partial_2 f) \quad \text{on } \mathcal{H}_c$$

Open question:

- Incorporating fault-tolerance or circuit-depth optimality
- Higher-order topological invariant
- Higher-form gauge fields

Recent developments

- Algebraic operation of chain complexes gives new family of QEC codes
 - Hypergraph product codes → tensor product [Tillich, Zémor 2014]
 - Fiber-bundle codes → fiber bundle [Hastings, Haah, O'Donnell 2021]
 - Lifted product codes → lifted product [Panteleev, Kalachev 2021]
- Inclusion of measurement error leads to longer chain complexes
 - Higher-dimensional QEC codes
 - Fault-complex [Hillmann et al. 2025]
 - Single-shot QEC [Campbell 2018]

Mathematical tools are useful to analyze these codes.

Recovering known transversality criteria

Known criterion	In this framework
Divisibility: a 2^m -divisible CSS code admits transversal $\exp(i\pi Z/2^m)$	$\beta_2^{(v)} = 0$ for uniform θ_v (the homology class lifts)
Tri-orthogonality $\Rightarrow$ transversal T	level $m = 3$ with $\beta_1^{(v)} = \beta_2^{(v)} = 0$ (uniform angle)
Quasi- / multi-level generalizations	non-uniform θ_v : the full classification

Divisibility and tri-orthogonality are the uniform-angle corner of a full homological classification (non-uniform angles give strictly more transversal gates).

Relation to high-energy physics?

- Transversal Pauli Z rotation gate at level $m =$ “ 2^{m-1} -root of Wilson loop”
- Transversal implementability $= \exists$ gauge-invariant 2^{m-1} -root of $W(\gamma)$
- Example: $m = 2 =$ transversal S (our running case)

$$\begin{array}{ll} \text{gauge field side} & \text{QEC operator side} \\ W(\gamma) = \exp\left(\oint_{\gamma} a\right) & \leftrightarrow Z(\gamma) = \prod_i (Z_i)^{\gamma_i} \\ W^{(2)}(\gamma^{(2)}) = \exp\left(\frac{1}{2} \oint_{\gamma^{(2)}} \tilde{a}\right) & \leftrightarrow S(\gamma^{(2)}) = \prod_{i=1}^n (S_i)^{\gamma_i^{(2)}} \\ \text{with } W(\gamma) = (W^{(2)}(\gamma^{(2)}))^2 & \end{array}$$

Our transversal S is the case $m = 2$: the **square root** of the Wilson loop.
 $m = 3$ (4th root) = transversal T on $[[15, 1, 3]] \rightarrow$ next.

[[15, 1, 3]] Reed-Muller: setup

- Setup: the [[15, 1, 3]] punctured Reed-Muller code. Physical $T^\dagger$ on every qubit $\rightarrow$ logical T .
- In this framework: uniform angle $\theta_i \equiv 1$, level $m = 3$, $\mathbb{Z}_8$ coefficients.
- **Tri-orthogonality** (every triple overlap of H^Z rows is even) $\Rightarrow \beta_1^{(\nu)}, \beta_2^{(\nu)}$ **vanish** for $\nu = 1, 2, 3$.

[[15, 1, 3]] Reed-Muller: transversal T

- $\Rightarrow$ A $\pi/4$ rotation of logical Z is transversal = transversal T .
- HEP reading: $m = 3$, so logical $\bar{T}$ is the $2^{m-1} = 4$ -th root of the Wilson loop:

$$\bar{T}(\gamma) = W(\gamma)^{1/4}$$

Textbook “tri-orthogonal $\Rightarrow$ transversal T ” = both obstructions vanish at uniform angle, level 3.

Non-Abelian code and quantum double model

- Non-Abelian gauge groups $\rightarrow$ non-Abelian anyons = key to topological QC
- Quantum double model (Kitaev 1997):
 - A lattice model with the quantum double of a finite group G
 - $G = \mathbb{Z}_2$: reduces to the toric code
 - G is non-Abelian: realizes non-Abelian anyons (e.g. $G = S_3, D_4$)

Non-Abelian anyons and universal computation

- Non-Abelian anyons: matrix-valued exchange statistics $\Rightarrow$ universal topological QC
- Non-Abelian CSS? CSS is $\mathbb{Z}_2$ -based $\rightarrow$ generalization is a hot topic

Extending the gauge-field / homology dictionary from $\mathbb{Z}_2$ to non-Abelian G is wide open, and these tools should transfer.